Introduction to Work & Energy

Name: **Date:**

A. Lesson & Previous Material Recap

1) What is energy? Energy is the ability to do________________________________

2) What is force? A force is a ________________________________ that can change an object's motion.

3) What is displacement? Displacement refers to ____.

4) Work occurs when a ______________________________ causes a ______________________________.

B. Energy Formulas Review

1) What form of energy causes motion?

2) What is the formula for calculating this energy?

3) What form of energy causes an object to potentially fall?

4) What is the formula for calculating this energy?

5) What form of energy causes an object to potentially move based on a spring?

6) What is the formula for calculating this energy?

7) What are the units for energy?

C. A New Formula

1) Work (Force Parallel to Motion)
When a force is applied in the same direction as motion, work is given by:

This formula applies when force and displacement are aligned.

2) Work (Force at an Angle)
When a force is applied at an angle, only part of the force does work:

This formula accounts for the component of force along the direction of motion.

3) Units of Work
The SI unit of work is ________________________.

D. Types of Work

Positive Work:

Occurs when force and displacement are in the ________________________ direction.

Negative Work:

Occurs when force and displacement are in ________________________ directions.

Zero Work:

Occurs when there is no displacement & the force is perpendicular to motion.

D. Work–Energy Connection

The work done on an object results in a change in its __________________________.

The Work–Energy Theorem:

What happens to kinetic energy when net work is positive?

What happens to kinetic energy when net work is negative?

E. Conceptual Work Scenarios

For each situation, determine whether the work done by the listed force is **positive, negative, or zero**.

1. A person pushes a box forward, and the box moves forward.

2. Friction acts on a sliding book.

3. A student carries a backpack horizontally at constant speed.

4. Gravity acts on a ball moving upward.

5. Gravity acts on a falling object.

F. Work Calculations (Parallel Forces)

Problem 1: A 40 N horizontal force pushes a box 5.0 m across a floor. There is no friction.

(a) What force is doing work?____

(b) Calculate the work done.

(c) Is the work positive, negative, or zero?

Problem 2: A 20 N friction force acts on a sliding crate that moves 6.0 m forward.

(a) Calculate the work done by friction.

(b) Is the work positive, negative, or zero?

G. Work Calculations (Angled Forces)

Problem 1: A 50 N force pulls a sled 8.0 m at an angle of 30° above the horizontal.

(a) Identify the angle between force and displacement.

(b) Calculate the work done.

(c) Is the work positive, negative, or zero?

(d) Is the work less than, equal to, or greater than Fd?

Problem 2: A 120 N force lifts a box straight up 2.0 m.

(a) What is the angle between force and displacement?

(b) Calculate the work done.

H. Work–Energy Applications

Problem 1: A 2.0 kg cart initially at rest experiences 10 J of net work.

(a) What is the cart's final kinetic energy?

(b) What is its final speed?

Problem 2: A 1.5 kg ball moving at 6.0 m/s is stopped by friction.

(a) How much work must friction do?

(b) Explain the sign of the work.